Civil Engineering

Vectors, Matrices and Engineering Data Analysis

Trainer Guide - Keep Private

This guide gives marking expectations. Do not upload publicly if learners should not access marking guidance.

Marking Guidance

- Formula selection: 1–2 marks.
- Correct substitution: 1–2 marks.
- Accurate computation: 2–4 marks.
- Correct units: 1 mark.
- Technical interpretation: 1–2 marks.

Expected Response Pattern

1. For forces, the learner should identify the known values, select a valid formula, substitute correctly, compute accurately, and explain the answer in civil engineering context.

2. For survey displacement, the learner should identify the known values, select a valid formula, substitute correctly, compute accurately, and explain the answer in civil engineering context.

3. For matrix equations, the learner should identify the known values, select a valid formula, substitute correctly, compute accurately, and explain the answer in civil engineering context.

4. For quality control, the learner should identify the known values, select a valid formula, substitute correctly, compute accurately, and explain the answer in civil engineering context.

5. For statistical judgement, the learner should identify the known values, select a valid formula, substitute correctly, compute accurately, and explain the answer in civil engineering context.

References and Learning Sources

These learner materials are original notes prepared for Mr Mwirigi Mathematics Training Hub. The topic sequencing and level are informed by standard engineering mathematics references and the uploaded curriculum/OS materials. No textbook pages or solution-manual pages have been reproduced.

1. John Bird, Engineering Mathematics, Newnes / Elsevier.

2. K. A. Stroud and Dexter J. Booth, Engineering Mathematics, Palgrave Macmillan.

3. Uploaded KNEC solved mathematics materials for examination-style orientation.

4. Civil Engineering, Building Technology and Land Survey curriculum and occupational-standard materials used for local alignment.